FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER –1

TEST DATE: 20-12-2020

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks**: –2** In all other cases.

Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

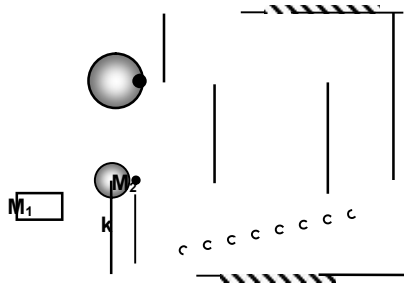
Physics

PART – I

SECTION – A
(MCQ Single Correct type)

This section contains 0**6** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Find the period of the free oscillations of M_1 if mass M_1 is slightly pulled down and released. Force constant of the spring is k, fixed pulley is massless and the smooth movable pulley has mass M_2.



(A) $T = 2\pi\sqrt{\dfrac{M_1 + M_2}{k}}$

(B) $T = 2\pi\sqrt{\dfrac{M_1 + 4M_2}{k}}$

(C) $T = 2\pi\sqrt{\dfrac{4M_1 + M_2}{k}}$

(D) $T = 2\pi\sqrt{\dfrac{3M_1 + M_2}{k}}$

2. A wire of length ℓ having tension T and radius r vibrates with a fundamental frequency f. Another wire of the same metal with length 2ℓ having tension 2T and radius $2r$ will vibrate with a fundamental frequency

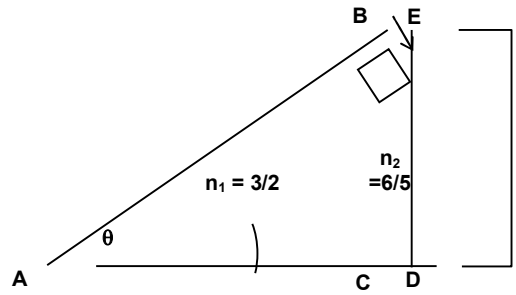
(A) f

(B) $\dfrac{f}{2\sqrt{2}}$

(C) 2f

(D) $\dfrac{f}{\sqrt{2}}$

3. In the figure shown ABC is the cross section of a right-angled prism. BCDE is the cross section of a glass slab. Find the value of θ so that the light incident normally on the face AB does not cross the face BC. [Given $\sin^{-1}(3/5) = 37^o$]



(A) $\theta < 37^o$

(B) $\theta > 37^o$

(C) $\theta \le 53^o$

(D) $\theta < 53^o$

4. The activity of a sample of radioactive material is R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. If mean life of the radioactive sample is T, then

(A) $R_1 t_1 = R_2 t_2$

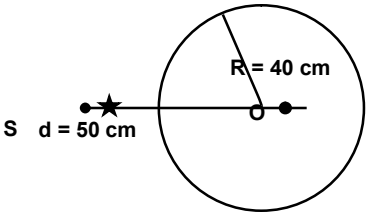
(B) $\dfrac{R_1 - R_2}{t_2 - t_1}$ = constant

(C) $R_2 = R_1 \exp\left(\dfrac{t_1 - t_2}{T}\right)$

(D) $R_2 = R_1 \exp\left(\dfrac{t_1}{Tt_2}\right)$

5. Star initially has 10^{40} deuterons. It produces energy via processes ${}_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + p$ and ${}_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + n$. If the average power radiated by the star is 10^{16}Watt, the deuteron supply of the star is exhausted in a time of the order of:
(mass of H^2 = 2.014u, m(p) = 1.007u, m(n) = 1.008u , m(He^4) = 4.001 u)
(A) 10^6 s
(B) 10^8 s
(C) 10^{12} s
(D) 10^{16} s

6. A point source S of light of power 2×10^{-3} Watt emitting mono energetic photons of energy 5eV each is placed at a distance 50 cm from the centre of an uncharged stationary metallic sphere of radius 40 cm and work function 2eV. The efficiency of photoelectron emission is one for every 10^6 incident photons. Assume that the sphere is isolated and the photoelectrons are instantly swept away after emission. The time after which the photoelectron emission stops will be



(A) 4.33 sec
(B) 2.33 sec
(C) 2.67 sec
(D) 1.67 sec

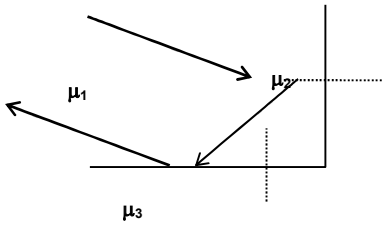
(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. In a resonance tube experiment, a closed organ pipe of length 120 cm resonates when tuned with a tuning fork of frequency 340 Hz. If water is poured in the pipe then choose the correct option(s). (Given v_{air} = 340 m/s)
(A) Minimum length of water column in the pipe to create the resonance is 45 cm
(B) The distance between the two successive nodes is 50 cm
(C) Maximum length of water column in the pipe to create the resonance is 95 cm
(D) Maximum length of water column in the pipe to create the resonance is 105 cm

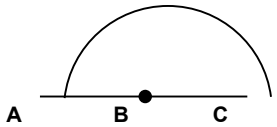
8. An ideal spring of natural length 40 cm has a spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to ceiling. The block is released from rest from the position, where the spring has length 45 cm. (g = 10 ms^{-2})
(A) The block will perform SHM of amplitude 5 cm.
(B) The block will have maximum velocity 30 $\sqrt{5}$ cm/sec
(C) The block will have maximum acceleration 15 m/s^2
(D) The minimum potential energy of the spring will be zero

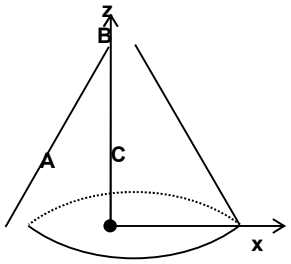
9. In the diagram shown, a ray of light is incident on the interface between 1 and 2 at angle slightly greater than critical angle. The light suffers total internal reflection at this interface. After that the light ray falls at the interface of 1and 3, and again it suffers total internal reflection. Which of the following relations should hold true?



(A) $\mu_1 < \mu_2 < \mu_3$
(B) $\mu_1^2 - \mu_2^2 > \mu_3^2$
(C) $\mu_1^2 - \mu_3^2 > \mu_2^2$
(D) $\mu_1^2 + \mu_2^2 > \mu_3^2$

10. A semi-circular silver sheet of radius R and mass m is folded to form a cone with its base in x-y plane as shown in the figure. If a block of mass m is sliding on its inclined surface with a constant velocity and a ray of light in x-z plane propagating along negative x-axis is incident on it, then which of the following is/are correct.





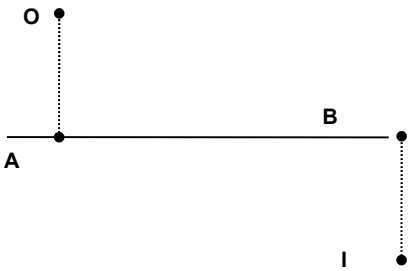
(A) $\mu_k = \sqrt{3}$

(B) $\mu_k = \frac{1}{\sqrt{3}}$

(C) unit vector along the reflected ray will be $\frac{\hat{i}}{2} + \frac{\sqrt{3}}{2}\hat{k}$

(D) unit vector along the reflected ray will be $\frac{\hat{i}}{2} + \frac{\sqrt{3}}{2}\hat{j}$

11. A luminous point object is placed at O, whose image is formed at I as shown in the figure. Line AB is the optical axis. Which of the following statement is/are correct?



(A) If a lens is used to obtain the image, then it must be a converging lens and its optical center will be the intersection point of line AB and OI.

(B) If a lens is used to obtain the image, then it must be a diverging lens and its optical center will be the intersection point of line AB and OI.

(C) If a mirror is used to obtain the image then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.

(D) I is a real Image.

12. Points A (0, 1cm) and B (12cm, 5cm) are object image pair (one of the point acts as object and the other point as image) x-axis is the principal axis of the mirror. Then this object image pair is:

(A) Due to a convex mirror of focal length 2.5 cm

(B) Due to a concave mirror having its pole at (2 cm, 0)

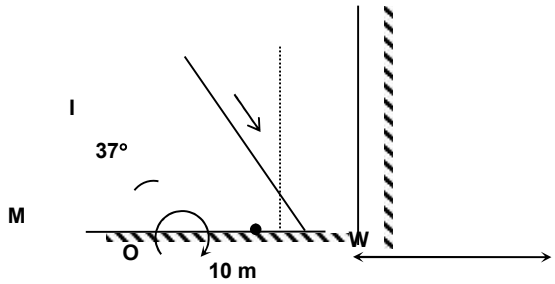
(C) Real virtual pair

(D) Data is insufficient for (A) and (B)

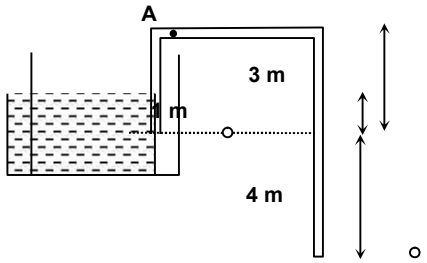
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A light ray I is incident on a plane mirror M. The mirror is rotated in the clockwise direction at a frequency $\frac{9}{\pi}$ rev/sec. The light reflected by the mirror is received on the wall W at a distance 10 m from the axis of rotation. When the angle of incidence becomes 37° and the plane mirror is horizontal as shown in the figure. Find the speed (in m/s) of the spot (a point) on the wall.



14. A liquid of specific gravity 0.5 is filled in a large container shown in the figure. Area of cross section of tube is negligible as compared to the area of cross section of container. If the pressure at point A is given by $x \times 10^4$ N/m^2. Find x. (Given atmospheric pressure, $P_0 = 10^5$ N/m^2, g = 10 m/s^2)



15. In YDSE, a thin plate of glass of refractive index 1.4 is placed normally in the path of one of the coherent interfering beams of a monochromatic light of wavelength 5000 Å. If central bright band of fringe system is formed at the position of second bright band from centre, when no plate is placed. Find the thickness of the glass plate in μm.

16. A point source of power $\left(\frac{9}{2}\right)\pi$ watts is producing sound waves. The velocity of sound is 330m/s, density of air is 1.0 kgm^{-3}. Then find the pressure amplitude (in Nm^{-2}) at a distance r = $\sqrt{330}$ m from the point source.

17. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz. The frequency emitted when it vibrates in the just next higher mode such that this point is again a point of maximum displacement is η × 40 Hz. Find the value of η.

18. In a setup of displacement method experiment, distance between the screen and a light source is 140 cm and the lens used has a small aperture. By moving the lens between the source and the screen, sharp images are obtained on the screen for two different positions of the lens. The ratio of sizes of these two images is 16 : 1. Find the focal length of the lens (in cm).

Chemistry

PART – II

SECTION – A
(MCQ Single Correct type)

This section contains 0**6** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

19. The geometry of $Ni(CO)_4$ and $\left[Ni(PPh_3)_2Cl_2\right]$ are respectively:
(A) Both square planar
(B) Tetrahedral and square planar respectively
(C) Both tetrahedral
(D) Square planar and tetrahedral respectively

20. The chemical composition of 'slag' formed during smelting process in the extraction of copper is
(A) $Cu_2O + FeS$
(B) $CuSiO_3$
(C) $FeSiO_3$
(D) $CuFeS_2$

21. Which of the following reaction is taking place in the final step of metallurgical extraction of Cu metal from copper pyrite in Bessemer furnace?
(A) $4Cu_2O + FeS \longrightarrow 8Cu + FeSO_4$
(B) $Cu_2S + O_2 \longrightarrow 2Cu + SO_2$
(C) $Cu_2S + 2FeO \longrightarrow 2Cu + 2Fe + SO_2$
(D) $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

22. Which of the following compounds dissolves in a mixture of $NaOH/H_2O_2$ and undergoes oxidation?
(A) BeO
(B) $Cr(OH)_3$
(C) ZnO
(D) $Al(OH)_3$

23. Which of the following is correct match?
(A) Both $XeOF_4$ and BrF_5 contains same number of valence electrons and are square pyramidal
(B) Both $XeOF_4$ and XeO_6^{4-} contains same number of valence electrons and are octahedral
(C) Both XeO_6^{4-} and BrF_5 contains same number of valence electrons and are square pyramidal
(D) Both $XeOF_4$ and BrF_5 contains same number of valence electrons and are octahedral

24. A salt on treatment with dilute H_2SO_4 liberates a colourless gas, the gas gives following observations
(i) White turbidity with Baryta water
(ii) Turns acidified dichromate solution green
(iii) Gives yellowish white turbidity with H_2S
The anion in the salt is
(A) CO_3^{2-}

(B) $C_2O_4^{2-}$

(C) SO_3^{2-}

(D) NO_2^-

(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Coordination compounds have great importance in biological systems. Which of the following statement(s) is/are correct?
(A) Chlorophyll is a green pigments in plants and contain calcium
(B) Cyanocobalamin is vitamin-B_{12} and contains cobalt
(C) Carboxypeptidase-A is an enzyme and contains zinc
(D) Haemoglobin is the red pigment of blood and contains iron

26. A blue colouration or blue ppt. is obtained when
(A) NH_3 is mixed in $CuSO_4$ solution
(B) $FeCl_3$ reacts with $K_4\left[Fe(CN)_6\right]$
(C) $FeCl_2$ reacts with $K_3\left[Fe(CN)_6\right]$
(D) Alkali metal dissolve in NH_3 to form dilute solution

27. XeF_4 can act as
(A) Flouride ion donor
(B) Reducing agent
(C) Oxidising agent
(D) Flouride ion acceptor

28. A binary solution of liquid 'A' and 'B' will show negative deviation from Roult's law if it fulfills the following condition(s):
(A) $P_A > P_A^oX_A$ and $P_B > P_B^oX_B$
(B) The intermolecular forces of A – B > A – A and B – B
(C) ΔH of mixing is negative
(D) ΔS of mixing is positive

29. A solid mixture of NaCl and $K_2Cr_2O_7(s)$ is gently warmed with conc. H_2SO_4. Which of the following observation(s) is/are correct?
(A) Deep red vapour is evolved
(B) Chlorine gas is formed as one of the major product
(C) These deep red vapours on dissolving in NaOH and followed by the mixing of $AgNO_3$ in solution, a brick red ppt. obtained
(D) Yellow solution obtained on mixing red vapours in NaOH, gives yellow ppt. with $Pb(CH_3COO)_2$

30. If we dilute a solution containing non-volatile solute, on adding water. Which of the following observation is/are correct?
(A) Vapour pressure decreases
(B) Its osmotic pressure decreases
(C) Boiling point decreases
(D) Freezing point decreases

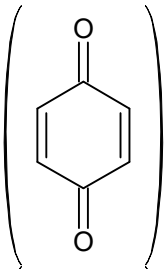
SECTION – C
(Numerical Answer Type)

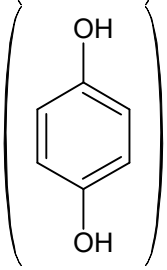
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. A substance form face-centred cubic crystals its density is 1.95 gm/cc and edge length of the unit cell is 625 pm. Calculate the molar mass of the substance (g/mole).

32. A solution containing 0.012 kg of $Ba(NO_3)_2$ in 0.1 kg of water boils at 100.57°C. Calculate the degree of ionization of the salt $K_b(\text{water}) = 0.52 \text{ K kg mol}^{-1}$ (Atomic mass of barium = 137.3)

33. Determine the magnitude of the EMF of the following cell at 298 K if pH = 6.0
$Pt \mid Q, QH_2, H^+ \parallel KCl(1\,M) \mid Hg_2Cl_2(s) \mid Hg(1) \mid Pt$
(Answer must be given in mili volt)

Where Q is quinone 

Where QH_2 

$E^\circ_{Cl^-|Hg_2Cl_2|Hg} = 0.280$ V and $E^\circ_{Q,QH_2,H^+|Pt} = 0.6996$ V

34. Henry's law constants as defined by the relation $K_H = P_A / X_A$ of oxygen and nitrogen dissolved in water at 273 K are 2.53×10^9 Pa and 5.47×10^9 Pa, respectively. A sample of water at a temperature above 273 K was equilibrated with air $(18\%\ O_2 \text{ and } 82\%\ N_2)$ at 1 atm. The dissolved gas was separated from a sample of this water and then dried. Calculate the % of N_2 in this dried sample $(1 \text{ atm} = 1.01 \times 10^5 \text{ Pa})$

35. 0.5 mol of an ideal gas 'A' is present in a chamber and 0.5 mol an ideal gas 'B' is present in an other identical chamber. Both the chambers are connected through a valve. Both the gases are at 1 atm pressure and same temperature. What will be the total change in entropy (in J/K) when valve is opened between the chambers?

36. $Pt(s) \mid H_2(g)(1\,\text{atm}) \mid HCl(aq) \mid AgCl(s) \mid Ag(s)$
if the cell potential for HCl is 493 mv, What is the pH of the solution in which the electrodes are immersed? $(E^\circ_{cell} = 0.222)$

Mathematics

PART – III

SECTION – A
(MCQ Single Correct type)

This section contains 0**6** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

37. Let $\bar{a}, \bar{b}, \bar{c}$ are three non coplanar vectors such that $3(\bar{a}\times\bar{b})+5(\bar{b}\times\bar{c})=\bar{a}$ and $\bar{d}$ is the unit vector coplanar with $\bar{b}$ and $\bar{a}\times\bar{b}$, then value of $\dfrac{(2\bar{a}+\bar{c}+3\bar{d})\cdot\bar{a}}{[\bar{a}\ \bar{b}\ \bar{c}]}$ is

(A) 5
(B) –5
(C) 13
(D) 8

38. From a moving point P lying on the plane $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=p$, perpendiculars PA, PB, PC are drawn to coordinate planes. The locus of the foot of perpendicular drown from origin to plane passing through points A, B and C is

(A) $\left(x^2+y^2+z^2\right)\left(\dfrac{1}{ax}+\dfrac{1}{by}+\dfrac{1}{cz}\right)=p$

(B) $\left(x^2+y^2+z^2\right)\left(\dfrac{1}{ax}+\dfrac{1}{by}+\dfrac{1}{cz}\right)=2p$

(C) $\left(\dfrac{1}{x^2}+\dfrac{1}{y^2}+\dfrac{1}{z^2}\right)\left(\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}\right)=p$

(D) $\left(x^2+y^2+z^2\right)\left(\dfrac{a}{x}+\dfrac{b}{y}+\dfrac{c}{z}\right)=p$

39. If a, b, c are three distinct positive real numbers, then system of equation (b + c – a)x – cy = 0
abx – (c + a – b)(a + b – c)y = 0 has
(A) trivial solution
(B) infinitely many solution
(C) no solution
(D) trivial or non trivial solution depending upon values of a, b, c

40. A person tosses a fair coin n times, he wins if he is able to get heads in multiples of 3. The probability he wins given that he gets head atleast one time is (n is not multiple of 3 and $n \geq 4$)

(A) $\dfrac{1}{3}$

(B) $\dfrac{2^n-4}{3\left(2^n-1\right)}$

(C) $\dfrac{2^n-3+(-1)^{n+1}}{3\left(2^n-1\right)}$

(D) none of these

41. The vertices A, B, C of ΔABC are 0, 3 and 4i on the argand plane. Three equilateral triangles (ΔPBC, ΔQAC and ΔRBA) are constructed out wardly to given ΔABC. The distance between centroid of ΔPQR and orthocentre of ΔABC is

(A) $\frac{5}{3}$

(B) 5

(C) 4

(D) $\frac{5}{6}$

42. If $T_1, T_2, ..., T_{100}$ are the distinct terms in Harmonic progression and T_6, T_{10}, T_{18} are in Arithmetic progression. If T_p, T_{12}, T_q are in Geometric progression, then maximum value of q is ($p < 12 < q$)

(A) 30

(B) 48

(C) 66

(D) 75

(One or More than one correct type)

This section contains **06** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If $x + y + z = 0$ and $x^3y + y^3z + z^3x = -9$, then choose the correct option(s)

(A) for $x, y, z \in R$, $xy + yz + zx = -3$

(B) for $x, y, z \in R$, $xy + yz + zx = 3$

(C) for $x, y, z \in C$ (complex number), $xy + yz + zx = \pm 3$

(D) if x, y, z are not all distinct, then $(|x| - 1)(|y| - 1)(|z| - 1) = 0$

44. Let $n = 2^3 3^4 7^2 11^5$. Choose the correct option(s)

(A) The number of divisors of n, which are of the form $6k + 4$, is 27

(B) The number of divisors of n, which are perfect squares is 36

(C) The number of divisors of n^2, which are not divisors of n, is 3105

(D) The number of divisors of n^2, which are not divisors of n but less n is 1373

45. Let $M = \left[a_{ij}\right]_{3\times 2}$ and $N = \left[b_{ij}\right]_{2\times 3}$ be two matrices such that $(MN)^2 = 3MN$ and $\det(NM) \neq 0$, then choose the correct option(s) (P is 2×2 matrix)

(A) det(NM) = 9

(B) if P(NM) = I, then $\lim\limits_{n\to\infty} \det\left(P + P^2 + P^3 + ... p^n\right) = 9$

(C) det(NM) = 27

(D) if P(NM) = I, then $\lim\limits_{n\to\infty} \det\left(P + P^2 + P^3 + ... p^n\right) = \frac{1}{4}$

46. A six digit number is formed randomly using digits {1, 2, 3} with repetitions. Choose the correct option(s)

(A) Probability that all digits are used at least once is $\frac{20}{27}$

(B) Probability that digit 1 is used odd number of times and 2 in used even number of times is $\frac{1-3^{-6}}{4}$

(C) Probability that all digits are used as well as odd digits are used odd number of times and even digit is used even number of times is $\left(\frac{3^6 - 2^7 + 1}{4 \cdot 3^6}\right)$

(D) Probability that exactly two digits are used is $\frac{62}{243}$

47. Let $f(n) = {}^{n}C_0\ {}^{3n}C_n - {}^{n}C_1\ {}^{3n-3}C_n + {}^{n}C_2\ {}^{3n-6}C_n$ ..... choose the correct option(s)
(A) Remainder when f(f(f(n))) is divided by 7 is 6
(B) Remainder when f(f(n)) is divided by 5 is 3, given n is even
(C) Remainder when f(f(n)) is divided by 5 is 2, given n is odd
(D) Remainder when f(n) is divided by 2 is 1

48. Let a, b, c, d are non zero real number such that a and 2b are roots of $x^2 - 9cx - 4d = 0$ and c and 2d are roots of $x^2 - 9ax - 4b = 0$. Choose the option(s) which can be true?
(A) a and c are the roots of the equation $x^2 + 4x + 4 = 0$
(B) a and c are the roots of the equation $x^2 - 20x + 4 = 0$
(C) If a and c are distinct then $a^2 + c^2 = 392$
(D) If a and c are equal then $b + d = -16$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If the minimum value of $(a^3x^2 + b^3y^2 + c^3z^2)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)^2$ is 216. (a, b, c, x, y, z $\in R^+$), then find the minimum value of $a^2 + b^2 + c^2$.

50. Let z satisfies $z^3 = \alpha^3 z_1^3 + (1-\alpha)^3 z_2^3 + 3(\alpha - \alpha^2) z z_1 z_2$ and $f : R \to [0, \infty)$, $f(\alpha) = |z|^2$, find minimum value of $f(\alpha)$ given that $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$

51. The planes $\bar{r}\cdot\bar{x}_1 = 8$, $\bar{r}\cdot\bar{x}_2 = 5$ and $\bar{r}\cdot\bar{x}_3 = 3$ have a common line of intersection. Find the value of $\left[\bar{x}_1\ \ \bar{x}_2\ \ \bar{x}_3\right] + \left|(\bar{x}_1\times\bar{x}_2)\times(\bar{x}_3\times\bar{x}_2)\right|^2 + \left|8(\bar{x}_2\times\bar{x}_3) + 5(\bar{x}_3\times\bar{x}_1) + 3(\bar{x}_1\times\bar{x}_2)\right|^2$

52. Let $a_i = 1 \quad 1 \le i \le 10$
$2 \quad 11 \le i \le 20$, then sum $\sum\sum\sum_{1\le i<j\le k\le 20} a_i a_j a_k$ is equal to

53. A sequence $< x_n >$, $n \ge 0$, $n \in I$ be given by $x_{n+1} = \frac{1}{3}\left(x_{n-1} + \frac{1}{x_n}\right)$, $n \ge 1$. If the sequence is periodic find the value of $x_0 \cdot x_1$

54. Let $A = \left[a_{ij}\right]_{n\times n}$ such that $a_{ij} = \frac{(-1)^i(2i^2+1)}{4j^4+1}$, then value of $1 + \lim_{n\to\infty}\left(\text{trace}\left(A^n\right)\right)^{\frac{1}{n}}$ is